LIBBY — PATIENT/CAREGIVER TRACK

aml-mds-related-rr-tp53-aberrant-hla-pending-x7q2-rerun2

Plain-language summary for patient and caregiver

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This is decision-support information, not a treatment plan. Please share it with your oncologist before making any decisions based on what's here.

What this page is

Libby is a research tool. It takes the specific features of one person's cancer, gathers the treatments built to aim at those features, and lays out what the published evidence actually says about each one. This page is the plain-language version of that work: what we understand about your leukemia, what you told us mattered, the tests that decide several of the answers, and then the options in order, with the upside and the catch of each written as plainly as we can manage.

You came here with a specific question about HLA-matched immune-cell therapy, so here is the short answer up front. That path is narrower than you were probably hoping, and it is not closed. The reason it is still open is genuinely counterintuitive, and it is explained in Option 3.

What we know about your cancer

You have acute myeloid leukemia that grew out of an earlier bone-marrow disorder. It came back about six years after a stem-cell transplant from your matched sibling, which is a long remission by any standard. The pattern points to the leukemia finding a way around your donor's immune cells rather than the graft failing.

Where things stand now shapes almost everything below.

The marrow is very full. Roughly 85 out of every 100 cells in your bone marrow are leukemia cells, and the count in your blood climbed steeply over a couple of weeks. There is also a solid deposit of leukemia in your right thigh, which doctors read as an aggressive feature and which needs its own local plan, since nothing aimed at the marrow will necessarily treat it.

The last chemotherapy did not work. A strong reinduction regimen was given and the leukemia grew through it, including through a drug called venetoclax. That matters later, in Option 10.

Your leukemia cells carry two surface proteins that drugs can grab onto, called CD33 and CD123. Both were confirmed present. What nobody has measured is how much of each protein sits on the cells, and the amount is what predicts whether those drugs do anything. One of the trials below will not even screen you without that number.

The tissue-type result the whole immunotherapy question depends on has not been retrieved. High-resolution HLA typing was done when you were worked up for your first transplant. The detailed result exists somewhere in that record, but it was never copied into the chart your current team is reading, where it sits as a placeholder. More on that below, because it is the first thing to fix.

A gene called TP53 is genuinely unsettled. A test at diagnosis found a damaging change in it. A more recent test did not find it, but that test used a panel that could easily have missed it: it cannot see missing copies of the gene at all, and it only detects a change if enough cells carry it. So this is not resolved in either direction, and nobody should tell you it is. It is being carried as probably abnormal, pending a better test.

The chromosome picture is complex and higher risk. That is part of why the plan below leans toward a transplant-based route with curative intent rather than something gentler.

You are up and about and looking after yourself. One real gap: there are no recent heart, kidney, or liver numbers on file. Several options below cannot be dosed safely, or even chosen between, until that fitness check is done.

What you told us matters most

You want a shot at cure. Not holding the line. The ranking is built around that.

You lean toward the stronger treatment and will accept more side effects to get it. That lean is provisional, and you asked for it to be revisited once your fitness is documented, which is one reason the heart and organ labs sit in the first step.

You are open to clinical trials, including traveling for them. Most of what follows lives inside trials, so they are surfaced where they fit rather than saved for last.

A second transplant and cell therapy are both acceptable to you. You ruled nothing out. That matters, because the route with the only documented cure fraction runs through a second transplant.

And the explicit reason you came: HLA-matched immune-cell therapy, the kind that targets HA-1 and HA-2. De-risking that pathway was the priority of this whole run.

The first step everyone agreed on

The single most useful thing anyone can do this week is not a drug. It is a records request and a phone call.

Your high-resolution HLA typing already exists. It was run when you were prepared for your first transplant and it is sitting in that transplant center's file or in the donor registry record. Retrieving it takes days. Re-running it fresh takes three to seven days and can be rushed. Either way, ask for the whole class I result to be transcribed, not just the one line, because a second marker on the same page (HLA-A*24:02) is the only backup thread if the main one is negative.

Alongside that, four other things belong on the same requisition.

A genetic test called HA-1/HA-2 genotyping, run on you and on your sibling donor. HA-1 and HA-2 are tiny differences that even a perfectly matched sibling can differ on, and the therapy in Option 3 only works if you and your sibling differ in one specific direction. This is the slow item. It is a specialized send-out that runs one to three weeks, it needs a fresh blood sample from your sibling, and it needs your sibling's consent, which nobody has asked for yet. Nothing can be manufactured until it comes back, so it should go out the same day as everything else rather than after a response is documented.

A quantitative measurement of how much CD33 and CD123 sit on your leukemia cells. Not "present or absent," which is what you already have, but a number. One trial below screens on CD33 being on at least 20% of blasts.

A better TP53 test: retrieving the original result from diagnosis (including which exact change it was), plus a stain and a chromosome test on stored tissue, plus a sequencing panel that can see missing copies. This is what resolves the contradiction without collapsing it in either direction.

Heart, kidney, and liver function. Every statement about side effects on this page is currently being made without those numbers.

Two things about this step deserve saying plainly. First, none of it hurts. It is blood draws, a marrow sample already indicated, stored tissue, and a phone call. The only cost is time. Second, and this is important: **this runs at the same time as starting treatment, not before it.** Your blood counts moved fast. Nothing here justifies waiting for a lab result before treating. Step one means send it today, not hold everything until it comes back.

The options

Ten options came out of the analysis, in the order below. The tests above are the true first move; the numbering here covers the treatments. The first two are the core of the plan. Options 3 through 9 each carry a specific caution, spelled out. Option 10 is one Libby is actively recommending against, and it is on the page so you can see why rather than wonder.

One thread runs through all of it. With the marrow this full, most of the immune-based options below simply cannot work yet, and several of them will not even enroll you. One trial caps marrow leukemia at 30%, another at 10%, another wants your white count under 25,000. The engineered T-cell approaches have repeatedly disappointed when there is this much disease to chew through. That is sequencing, not exclusion. The leukemia has to come down first for the interesting things to become possible at all.

Option 1 — CLAG-GO (chemotherapy plus a CD33-targeted antibody drug), on a trial

This is a chemotherapy combination: three drugs given together, plus an antibody called gemtuzumab that uses the CD33 handle on your leukemia cells to deliver a toxin inside them. The antibody is given in three small split doses rather than one big one, which is the schedule with the better liver-safety record.

Its job here is to empty the marrow so that everything else becomes reachable. It is the one strongly fitting open trial in the whole search that treats relapse after a transplant as eligible rather than disqualifying, and it accepts people whose disease was refractory to an intensive regimen, which yours was.

What it might do is hard to state precisely, and I would rather say so than invent a number. The combination itself has no published results in refractory leukemia at all. The closest evidence is the antibody used alone in 57 people at first relapse, where about 26 out of 100 reached a complete remission, and seven of those people went on to a transplant afterwards. Your situation is harder than theirs, so treat that as a ceiling rather than an expectation.

There is a specific honesty point about the antibody component. The strongest evidence in this entire file pooled more than 3,000 patients from randomized trials and found essentially no survival benefit from that antibody in people whose chromosome pattern looks like yours. The difference was about two people in a hundred at six years, which is not a difference anyone can rely on. The chemotherapy backbone is what earns this the top spot. The antibody component is a reasonable addition with a good liver-safety record before a transplant, not the reason it works.

The side effects are heavy. Expect weeks of very low blood counts (in the antibody study, about 23 days), fevers, and infections, in a marrow that has been carrying a donor graft for six years and may recover slowly. The antibody carries a risk of a serious liver complication where small veins in the liver clot off. Splitting the dose lowers that risk without erasing it, and another conditioning regimen may be coming within months, so baseline liver labs before the first dose and weekly checks afterwards are the condition. There is also a real chance this simply does not work, because you progressed through a closely related chemotherapy drug already.

This matches your curative goal and your openness to trials. If the trial slot does not open, this regimen can be rebuilt from marketed drugs at your own center, which is a fallback almost nothing else here has.

Option 2 — A second transplant from your sibling, once the leukemia is reduced

This is the only option on this page with a documented cure fraction. That sentence is doing a lot of work, so here is what is behind it, including the parts that are missing.

Four separate registries, covering thousands of people who relapsed after a transplant and went on to a second one, point in the same direction. In the group most like you, where the first donor was a sibling, about 37 out of 100 were alive two years later. Across a much larger registry of second transplants, about 15 out of 100 were alive without relapse at five years. You sit on the favorable side of nearly everything those registries measure: a sibling donor who has already donated once, a six-year first remission, which is long.

The exception is the one that decides the timing, and it is not a small one. A registry that scored people by how much leukemia they carried going into the second transplant found that those in the worst band, which is where you are today, had about 4 out of 100 alive at two years. Taking you to a second transplant with the marrow as it stands would spend all of the risk and buy very little of the benefit. Get the leukemia below roughly a quarter of the marrow and the favorable numbers start applying to you.

Two honest limits on those numbers. They are all backward-looking, and none of them adjusts for the obvious fact that the people who were offered a second transplant were the ones their doctors judged fit enough for it, so the survival figures flatter the treatment. And not one of those registries reports how many people died from the transplant itself rather than from leukemia, so the harm side of every curve quoted above is missing.

The risks are the ones you already know from the first time, and they are larger the second time: graft-versus-host disease, months of low counts, infection, and death from the procedure itself. How intense the conditioning should be cannot be decided without a measured heart and kidney baseline, which is another reason those labs are in the first step. And a marrow-directed transplant does not treat the thigh deposit, which needs its own plan.

One unresolved question sits underneath this. Your leukemia already escaped your sibling's immune cells once. Giving the same donor's cells again does not, by itself, fix whatever let that happen. Testing the relapsed leukemia cells for whether they have hidden the markers the donor cells recognize is worth doing before committing.

Option 3 — Engineered T cells that target HA-1

This option is only available if the HLA typing comes back positive for HLA-A*02:01, and if you and your sibling differ at HA-1 in the specific required direction (you positive, your donor negative). If the typing comes back negative, this option is off the table entirely rather than reduced, and so is Option 5. That would leave no HLA-matched immune-cell option ranked anywhere on this page. The other options here aim at different features of your leukemia and are unaffected either way. There is one narrow thread left: if the main marker is negative but the second one on the same report (HLA-A*24:02) is positive, there is a different engineered-cell approach built on that marker, but no route currently exists that a patient in the United States can enter, which is exactly why both markers should be transcribed off the same page rather than one of them. Beyond this page, standard care for relapsed leukemia exists and is worth discussing, but it sits outside what Libby ranks. That is a separate conversation with your oncologist.

You came looking for HA-1/HA-2 T-cell therapy. What the search found is a reversal that almost nobody would predict. The commercially developed versions of this therapy, the ones that get written about, all exclude people who have already had an allogeneic transplant. You have had one. A European protocol written specifically for sibling-donor transplant recipients was withdrawn before it enrolled a single patient. So the products you were most likely to have read about are closed to you.

But one academic trial is open, and it is open because it **requires** a prior transplant. Its whole design is treating relapse after an allogeneic transplant, using T cells from your original donor, engineered to recognize HA-1. The very thing that shuts the other doors is the entry ticket to this one. That is the single most important sentence on this page.

What it does: HA-1 sits only on blood-lineage cells, including the leukemia, and not on the rest of the body. So engineered cells that hunt HA-1 go after the leukemia and your remaining marrow while mostly leaving other organs alone. Given after your donor's immune system is back in place, the intent is a durable, curative remission.

What the evidence is: nine people. Of those nine, four either reached a complete remission or held one, and one of them stayed in remission past two years. Across all nine there was no cytokine release syndrome and none of the neurological toxicity this class of therapy is known for, and no dose-limiting side effects. That safety read is genuinely clean and unusual.

This option carries caveats, and here is the honest tradeoff. The trial's registered purpose was to test whether the cells can be manufactured and whether they are safe, not whether they work. So "four of nine" is a descriptive observation, not a response rate, and the nine include both people whose remission the cells created and people whose remission they merely maintained. Nobody should quote it to you as a 44% chance. It is nine people in a study where everyone got the treatment and there was no comparison group. The safety is well supported. The effectiveness is promising and unproven, and that gap does not close even if your typing comes back perfect.

Two practical things. The cells are made from your sibling's cells, so your sibling has to agree to a fresh collection before anything can be manufactured, and that conversation has not started. And it runs at a single center, so travel plus the chemotherapy given before the infusion plus low counts needs a local backup plan written down before you go. Nobody in the published series had received a second transplant before their infusion, so how this interacts with Option 2 is unmeasured rather than reassuring.

Option 4 — An infusion of your sibling's immune cells (donor lymphocyte infusion)

This option carries caveats. Here is the tradeoff to weigh.

This is the simpler version of the same idea: instead of engineering anything, you receive unmodified immune cells from your sibling, hoping they attack the leukemia the way they did the first time. No conditioning chemotherapy, no manufacturing, and your donor has already donated once.

It has the only comparison in this whole file with an untreated control group. Among 399 people who relapsed after a transplant, about 21 out of 100 who received donor cells were alive at two years, against about 9 out of 100 who did not. That comparison has a well-known flaw worth knowing about: the people who received the infusion were the ones their doctors judged well enough to receive it, so some of that difference is selection rather than the treatment.

Within the group who did get it, the split is stark and it is about timing. Given to people in remission or with favorable chromosomes, about 56 out of 100 were alive at two years. Given to people with active disease or an empty marrow, about 15 out of 100. Your long first remission puts you on the good side of the strongest predictor. Your current marrow puts you firmly in the 15 group today. That is an argument for reducing the leukemia before an infusion, not against the infusion.

Two cautions. The benefit and the main harm are the same mechanism: donor cells attacking your tissues. Severe graft-versus-host disease would close the door on a second transplant and on any cell therapy afterwards. And not one of these registry studies grades side effects in any systematic way, so the harm side genuinely cannot be quantified for you from this evidence. There is also the same unanswered question as in Option 2: these are the same donor cells that already lost control of this leukemia once.

The trial version of this is currently shut to you, because it caps marrow leukemia at 30% and allows only one prior treatment after transplant, which your last regimen already used. The therapy itself remains available outside a trial. Where this option earns its place is as the route that keeps curative intent alive without a second conditioning regimen, if the fitness workup says your heart or organs cannot take one.

Option 5 — CBX-250, an engager drug that uses the same tissue-type marker

This option is only available if the HLA typing comes back positive for HLA-A*02:01. It does not need anything from your sibling. If the typing is negative, this closes at the same moment as Option 3, and the negative-result explanation in Option 3 applies here too.

This option carries caveats, and the caveat is large. This drug is designed to grab a leukemia marker with one end and pull a T cell onto it with the other. No human being has yet been treated with it in a reported study. There is no efficacy readout, and there is no published side-effect profile of any kind. The reasons it is on the page are eligibility and reach: it accepts relapse after a transplant, it runs at eleven sites in the US rather than one, and the company publishes a route for people who cannot get a trial slot.

Two things to weigh honestly. In a first-in-human dose study, you might be assigned a dose chosen to be safe rather than one high enough to work, and nobody can tell you which. And the marker this drug goes after is not one anyone has confirmed on your leukemia cells; the option rides your tissue type rather than a target measured in your disease. It is worth asking the sponsor directly whether they measure that target on study or whether the tissue type alone qualifies you. Enrollment also wants your white count under 25,000, so this becomes real only after Option 1 is working.

Option 6 — BMS-986497, a CD33-targeted trial drug

This option carries caveats. Here is the tradeoff to weigh.

This is another antibody that uses the CD33 handle, this one delivering a molecule that shuts down a process leukemia cells need to survive. There is no published human data of any kind: not effectiveness, not safety. It is on the page because its entry criteria are unusually open across the 47 trials searched. It names leukemia arising from a prior marrow disorder, it only requires CD33 to be detectable rather than above a threshold, and its posted criteria contain no cap on how much leukemia you carry and no bar on a prior transplant, which is rare enough to be worth verifying with the site rather than assuming.

Its real value is independence. Options 3 and 5 both close on the same afternoon if the typing comes back negative. This one does not care about your tissue type at all.

Two practical cautions. The trial has several arms, and one of them includes venetoclax, the drug you have already progressed through. Pushing for the single-drug arm matters, because being assigned to the venetoclax arm would spend a trial slot on a drug that has already failed you. And enrolling takes weeks that a marrow at 85% may not have, so this competes with Option 1 for time rather than running alongside it.

Option 7 — Pivekimab sunirine, a CD123-targeted antibody drug

This option carries caveats. Here is the tradeoff to weigh.

This drug uses the CD123 handle to deliver a toxin into leukemia cells. Of everything aimed at CD123 here, it is the only one with results in your own disease at your own stage rather than borrowed from a different cancer: in 29 people with relapsed or refractory leukemia, about 21 out of 100 responded and about 17 out of 100 had a complete or near-complete remission. With only 29 people, the true figure could plausibly sit anywhere from about 8 to about 40 out of 100, so treat it as a signal rather than a number.

The concern is the liver. In an earlier dose study, two people out of 68 developed the same serious liver-vein complication mentioned in Option 1. Both cases happened at doses four to ten times higher than the dose now used, and both reversed. That is a dose-level finding rather than a property of the drug, and it should not be used to strike CD123 off your list. It also should not be waved away, because the drug's label carries a warning about liver toxicity and because a conditioning regimen may be coming within months, and there is no pre-transplant safety record for this drug comparable to the one for the antibody in Option 1. The condition is baseline liver labs and weekly monitoring, not exclusion.

It is approved for a different, rarer blood cancer, so using it for your leukemia would be off-label, and the leukemia trial cohorts are closed to new patients. Practically, this is the fallback for reducing the leukemia if the CD33 measurement comes back low.

Option 8 — Tagraxofusp, another CD123-targeted drug

This option carries caveats. Here is the tradeoff to weigh.

This one also uses CD123, delivering a toxin by a different route. It has a real advantage in familiarity: it has been approved and in use for years, so hospitals have a worked-out routine for spotting and managing its signature side effect.

The number attached to it in most write-ups, roughly 72 out of 100 responding, comes from a different, rarer blood cancer and from people receiving it as their first treatment. It does not carry across to a marrow as full as yours, and presenting it as your expectation would be misleading. The nearest data in your own type of leukemia is about 69 out of 100 responding in previously untreated patients, which is also a different situation from yours. In your setting, honestly, nobody can give you a reliable figure.

The side effects are the reason this sits low. About 19 out of 100 people develop capillary leak syndrome, where fluid escapes from small blood vessels, and deaths from it have been reported. Most people get liver enzyme rises, and a majority get low blood protein. With no baseline liver, heart, or protein labs on file, this is precisely the drug that cannot be started until the fitness check is done, and the trial cohort has a hard blood-protein requirement. Prior venetoclax may also exclude you unless enough time has passed since your last dose.

Option 9 — Astatine-211 targeted conditioning before a transplant

This option carries caveats. Here is the tradeoff to weigh.

The idea behind this one speaks directly to the bind you are in. Instead of using chemotherapy to clear the marrow before a transplant, an antibody carries a radioactive particle directly to blood cells, delivering the radiation where it is needed and largely sparing the rest of the body. That means going into a transplant with active leukemia instead of having to reach remission first, which chemotherapy may not achieve here.

The principle was tested properly once, in a study of 153 people, using a related but different radioactive drug. About 17 out of 100 given targeted-radiation conditioning were in a deep, lasting remission six months later, against essentially none of the people on standard care. Two large caveats sit on that. First, people in that trial did not live longer overall, and the regulator declined to approve the drug on those results and asked for another study. Second, and more directly relevant, that trial excluded anyone who had already had a transplant. So the 17-in-100 figure cannot be quoted for you at all.

The version discussed here uses a different radioactive particle, and it has no published human results whatsoever. What is behind it is laboratory work in mice. What keeps it on the page rather than in a footnote is that it is the only targeted-conditioning protocol found that admits someone with a prior reduced-intensity transplant.

Two hard gates decide whether it is even real for you. The protocol wants a half-matched relative as the donor rather than the sibling who donated before, so whether such a relative exists and is willing is a phone call worth making now rather than a discovery in three months. And if the conditioning you received the first time was the fully intensive kind, you would be excluded, which means retrieving that record too. It also requires your circulating leukemia count to be brought down first.

Option 10 — Rezatapopt

This option was considered but not recommended. It is on the page because Libby shows what was weighed and set aside, not only what was kept. You may have come across it as "the TP53 drug," so the reason matters.

Rezatapopt is designed to repair one very specific damaged version of the TP53 protein, called Y220C. It is not a general TP53 drug. Two problems stack up.

The first is biological, and it is the decisive one. In the only laboratory study available, the drug did restore the TP53 protein's normal shape in leukemia cells, but the cells largely did not die unless a partner drug was added. The partner that worked best is venetoclax, which is the drug you just progressed through. So even a perfect test result would leave this option without the partner it appears to need.

The second is about the test itself. The trial requires a certified result showing the Y220C change above a specific detection threshold, and that threshold sits right at the limit of the same class of panel that recently reported your TP53 as negative. In other words, the assay could decide your eligibility before the biology does, in either direction. On top of that, the trial's backbone drug works slowly, which sits badly against a marrow at 85% and a goal of cure, and no peer-reviewed effectiveness figure exists for this drug in any blood cancer.

Declining it does not close the TP53 question. The better TP53 testing in the first step still happens, and if that testing does confirm a Y220C change, this becomes worth revisiting, though it would need a different partner drug than the one this trial uses. One thing to be clear about: the older, general-purpose TP53 drug is not the fallback here. It looked promising in an early single-group study and then failed when it was properly randomized.

Questions to ask your oncologist

- Can someone call the transplant center today and pull my original high-resolution HLA typing, and can the entire class I result be transcribed into my chart rather than just the one line?
- Can the HA-1/HA-2 genotyping on me and my sibling be sent this week, and can we start the consent conversation with my sibling now, given that nothing can be manufactured until that result is back?
- Are we starting treatment at the same time as sending these tests, rather than waiting for results? What would make you delay treatment, if anything?
- If the HLA-A*02:01 result comes back negative, what does that leave, and does the second marker checked on the same report open anything you could realistically get me into?
- What is the target for how far we need to reduce the leukemia before a second transplant becomes reasonable, and how will we know when we're there?
- For the trial regimen in Option 1, can we get the quantitative CD33 measurement the screening requires, and what liver monitoring would you put in place given the risk to the liver veins before a possible transplant?
- Given that a second transplant from the same sibling is the only option here with a documented cure fraction, can we also test whether my leukemia cells have hidden the markers my donor's immune cells recognize, since that is what may have let it escape the first time?
- What happens about the deposit in my thigh? Does it need its own treatment separate from whatever we do for the marrow?
- When can I get the heart, kidney, and liver testing done, and how would poor results change which of these options stays on the table?
- If a second transplant turns out to be too much for me, is an infusion of my sibling's immune cells after the leukemia is reduced a realistic alternative here?

Sources

PubMed (PMID): 17051246, 17909197, 22167752, 23471305, 23918951, 25008258, 26367221, 29051183, 29526774, 31018069, 38052038, 38423051, 38683966, 39298738, 39939820, 40437170, 40608889, 41671533

ClinicalTrials.gov (NCT): NCT03113643, NCT03326921, NCT03670966, NCT04050280, NCT04086264, NCT05473910, NCT06419634, NCT06529731, NCT06616636, NCT06704152, NCT06928662, NCT06994676